

P0 Indic Languages for Nemotron Parse

Goal: Training data to expand OCR / parse models into Indic languages.

Type: documents that can be used to train OCR / parse models.

Volume: ~250 pages per language (in the language's script) to start with. Eventually we may need thousands of pages.

Languages:

- Hindi
- Bengali
- Gujarati
- Odia
- Assamese
- Punjabi
- Marathi
- Urdu
- Tamil
- Telugu
- Malayalam
- Kannada

Questions

- Should the document content be entirely in the target language, or is a mix of English/other languages and the target language acceptable?
 - [A mix of the target language and english or other languages is acceptable.](#)
- Would publicly available or web-sourced content be acceptable, provided it meets your licensing and usage requirements?
 - [Yes.](#)
- Do you require only source documents, or do you also need OCR/parsing annotations such as extracted text, layout labels, table structures, reading order, or bounding boxes?
 - [We do require annotations as well. See details in Appendix.](#)
- Are there any preferred document domains (government, financial, educational, healthcare, forms, reports, newspapers, books, etc.)?
 - [Diverse sources are best.](#)

- What is the preferred delivery format?
 - Delivery format is flexible, as long as it contains the required information defined in the labeling requirements
 - If a page contains only images and no readable text in the target language, should it be included or excluded?
 - These can be excluded
 - Is there a preferred document length, or can the 250 pages be sourced from multiple documents of varying sizes?
 - Can be varying sizes.
 - Are there any specific requirements around document quality, scan resolution, handwritten content, or diversity of layouts that we should consider while curating the dataset?
 - Wide variety is preferred, digital documents, scanned documents of varying quality, books, website renders, etc.
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Appendix

Model training format

The training loader expects one page image plus either page-level text or block annotations.

Preferred block-level JSON fields:

JSON

```
{
  "block_boxes": [[x1, y1, x2, y2], ...],
  "block_classes": ["Text", "Title", ...],
  "block_text": ["recognized text for this box", ...]
}
```

The three arrays must have the same length. Index *i* in each array describes the same block.

Coordinate convention

Use absolute pixel coordinates in the original page image coordinate system:

None

[x1, y1, x2, y2]

where:

- x1 = left edge in pixels
- y1 = top edge in pixels
- x2 = right edge in pixels
- y2 = bottom edge in pixels

Rules:

- x2 must be greater than x1.
- y2 must be greater than y1.
- A box should cover the visible region of the block, not the whole page unless the whole page is a single block.

Annotation levels

Level 1: Image + full-page text

- Minimum useful OCR supervision.
- Provide all visible text in natural reading order.
- No boxes/classes required.
- This can train the model when it is prompted with no bboxes/no classes.
- Keep text as faithful as possible to the page.
- Preserve paragraphs and line/section boundaries where meaningful.
- Teaches reading but not structure

Level 2: Image + ordered text blocks + boxes

- Better than page-only text.
- Provide block_boxes and block_text.
- Use one block per paragraph, title, list item, table, formula, caption, header, footer, picture, etc.
- Blocks should be in reading order.

- Teaches reading + structure

Level 3: Image + boxes + classes + text

- Full annotation
- Provide `block_boxes`, `block_classes`, and `block_text` for every meaningful page element.

Level 4: Rich structured annotations

- Include table LaTeX/HTML, formula LaTeX, reading order, captions/footnotes, and figure/table relationships.

Class labels

Use exactly these class strings unless a new class is explicitly approved:

- Text
- Title
- Section-header
- List-item
- TOC
- Bibliography
- Footnote
- Page-header
- Page-footer
- Picture
- Formula
- Table
- Caption

Tables

Best target format for Table blocks is LaTeX:

None

```
\begin{tabular}{ccc}  
...  
\end{tabular}
```

Guidelines:

- Preserve row and column structure.
- Use `\multicolumn` and `\multirow` if needed.

Formulas

Best target format for Formula blocks is LaTeX.

- Display formulas should be class Formula.
- Inline formulas inside a paragraph may remain inline in the Text block unless there is a separate visible formula region.

Pictures and figures

- Use class Picture for non-text visual regions.
- Captions should be separate Caption blocks.
- If picture has readable text this should be annotated
- If it cannot be easily transcribed these can be marked for no-text-in-pic prompting

Reading order

Order blocks the way a person should read the page:

1. Title/header content
2. Main body columns in natural reading order
3. Captions near their associated table/figure when appropriate
4. Footnotes/footers after main body text unless visually/semantically tied earlier

For multi-column pages, read down the first column, then the next column, unless the layout clearly indicates another flow.

Example minimal page-level sample

JSON

```
{
  "image": "page_0001.jpg",
  "text": "Full page text in reading order..."
}
```

Example preferred block-level sample

JSON

```
{
  "image": "page_0001.jpg",
  "block_boxes": [
    [120, 80, 1540, 160],
    [140, 240, 1500, 420],
    [140, 480, 1500, 820]
  ],
  "block_classes": [
    "Title",
    "Text",
    "Table"
  ],
  "block_text": [
    "ભારતીય અર્થવ્યવસ્થા પર અહેવાલ",
    "આ પૃષ્ઠમાં મુખ્ય આર્થિક સૂચકોનું વર્ણન છે.",
    "\\begin{tabular}{cc} વર્ષ & વૃદ્ધિ \\\\ 2023 & 7.1\\% \\\\ 
    \\end{tabular}"
  ]
}
```